October 17, 2026

The Editors
Physical Review E
American Physical Society

Dear Editors,

I am pleased to submit the manuscript entitled **"An Exact Rational Invariant for Forman–Ricci Curvature on Chebyshev Toroidal Lattices"** for consideration for publication in *Physical Review E* as a Regular Article.

Summary of the work. The manuscript studies the Forman–Ricci curvature on three-dimensional toroidal lattices governed by the Chebyshev (ℓ^∞) metric, as a function of the integer interaction range $\lambda \geq 1$. The main results are:

1. **Exact closed-form formula.** We derive the exact rational expression for the mean curvature per edge,
$$\langle \kappa_e \rangle(\lambda) = -\frac{47\lambda^5 + 141\lambda^4 + 156\lambda^3 + 85\lambda^2 + 24\lambda + 3}{2(4\lambda^2 + 6\lambda + 3)},$$
valid for any torus size $L \geq 4\lambda$ in the no-aliasing regime.
2. **Unique dimensionless invariant.** We construct the invariant $\mathcal{I}(\lambda) = |\langle \kappa_e \rangle| \lambda^3 / k(\lambda)^2$ and prove its uniqueness under scaling, linearity, and combinatorial domain normalization axioms.
3. **Exact geometric limit.** We prove that in the dense limit $\lambda \to \infty$, the invariant converges to the exact rational number $\mathcal{I}_\infty = 47/512$, and we identify its geometric origin as an intrinsic integral constant of the ℓ^∞ metric on the three-torus, arising from the exact integration of overlapping ℓ^∞ unit balls.

Novelty. To the best of our knowledge, this is the **first exact closed-form curvature invariant derived for a three-dimensional lattice with extended interaction range**. While Forman–Ricci curvature has been extensively applied to complex networks, exact analytical results on regular spatial lattices with extended interaction ranges have remained scarce. The Chebyshev lattice is particularly suited to analytical treatment because its coordination geometry is a solid cube, whose intersection volume factorises over the three coordinate axes, allowing exact computations that would be intractable for other ℓ^p metrics.

Scope and honesty. The manuscript makes **no claim** of connection to quantum gravity, nor of convergence to smooth Riemannian geometry. The connection to quantum gravity is mentioned only as historical context, and the paper explicitly states its limitations in Sec. VI.C. This transparency is intentional: we believe that exact benchmarks in discrete geometry are valuable in their own right, without the need for overinterpretation.

Verification. All analytical results are independently validated by three complementary methods:

- **Symbolic verification** (SymPy): the algebraic difference between the exact discrete sum and the closed-form expression is exactly zero.
- **Numerical verification** (SciPy): the ℓ^∞ continuum integral yields $C_\kappa = -47/8$ and $\mathcal{I}_\infty = 47/512$.
- **Brute-force verification** (NetworkX): for $\lambda = 3$ and $L = 12$, the graph contains 1 728 nodes and 295 488 edges, and the numerical mean curvature agrees with the closed-form expression to machine precision (absolute difference 0).

The complete reproduction package (code, data, figures) is available as Supplemental Material and will be deposited in a public repository (Zenodo) upon acceptance.

Suitability for Physical Review E. The manuscript sits at the intersection of discrete geometry, complex networks, and statistical physics, which are core areas of interest for *Physical Review E*. The exact closed-form result, the geometric interpretation of the invariant 47/512, and the rigorous verification package make it suitable for the journal's readership.

Declaration. The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All code is released under a CC BY-SA 4.0 license. The author declares no competing financial interests.

Thank you for considering this manuscript for publication in *Physical Review*

E. I look forward to your response.

Sincerely,

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